

Python assignment

1. Load the **Customer data** sheet in Python. Set churned as y. Remove churned and customer_id from X.
2. Make a stratified 80/20 training and test split. Check how many customers canceled in each split.
3. Train these five models separately: KNN, Gaussian Naive Bayes, Decision Tree, Random Forest, and Logistic Regression.
4. Use a StandardScaler pipeline for KNN and Logistic Regression. Fit the scaler using training data only.
5. For each model, report test accuracy, precision, recall, F1-score, and a confusion matrix.
6. Compare the models: **Which catches the most customers who cancel? Which produces the fewest false alarms?** Would you choose the same model for both goals?
7. Inspect what each model learned. For example, compare K values, display a small decision tree, examine forest feature importance, and inspect logistic regression coefficients.
8. Write a short recommendation explaining which model you would use to identify customers for a retention campaign.